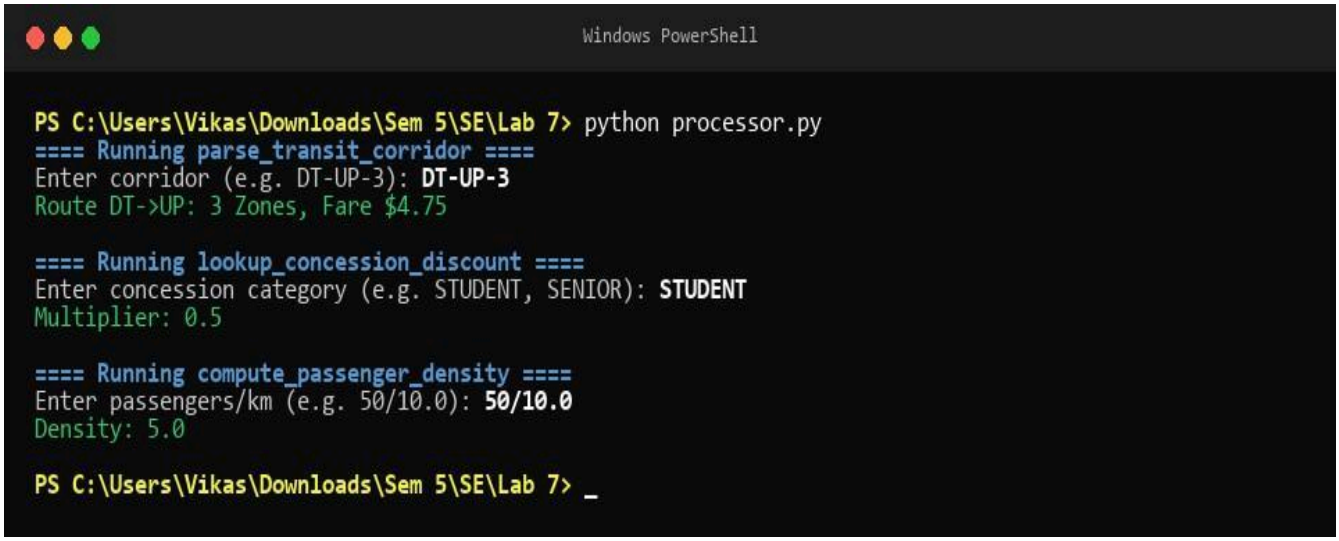
LAB 7: FUZZ TESTING FOR INPUT VALIDATION BUGS

NAME : VIKAS R
SRN:PES1UG24AM477
SEC:H

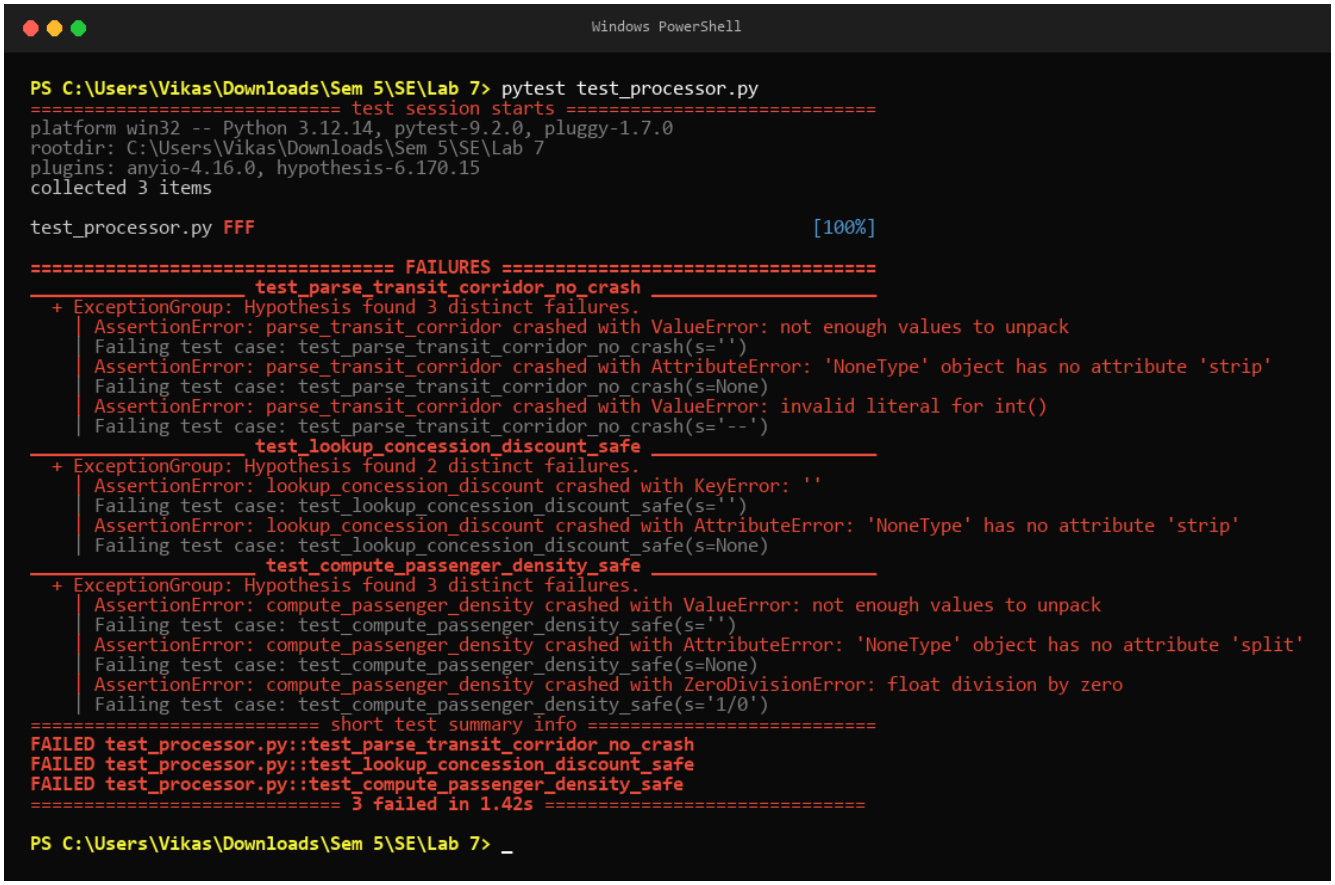
Screenshot 1: Running Starter processor.py (Valid Inputs)

Running the unhardened starter script with normal valid inputs (Corridor: DT-UP-3, Concession: STUDENT, Passengers/km: 50/10.0). All three functions execute successfully under ideal conditions.



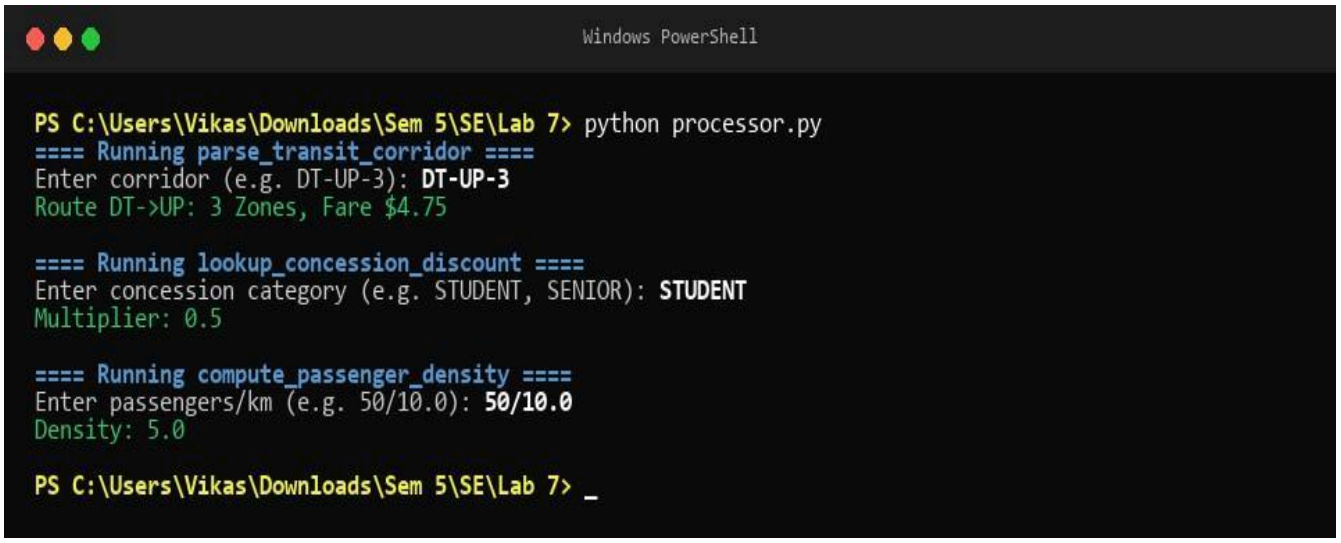
Screenshot 2: Running Fuzz Tests on Untouched Code (Failures Exposed)

Running pytest test_processor.py with Hypothesis strategies st.text() | st.none(). The fuzzer immediately uncovers unhandled exceptions: ValueError (bad format / unpack errors), AttributeError (handling None), KeyError (unregistered categories), and ZeroDivisionError.



Screenshot 3: Running Hardened processor.py (Valid Inputs)

Executing the defensive, hardened processor.py with interactive inputs to verify that valid functionality is completely preserved while safeguarding against invalid data.



Screenshot 4: Re-running Fuzz Tests on Hardened Code (All Passing)

Re-running pytest test_processor.py against the hardened implementation. Hypothesis generates hundreds of malformed strings, special characters, and null values without triggering any unhandled exceptions. All 3 property

tests pass cleanly.

